

Gains from Trade with Competitive Effects^{*}

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Abstract

This article examines the welfare gains from imports using a model with variable markups and multiple sectors within an input-output structure. Accounting for both pro-competitive and anti-competitive effects, I find that the surge of imports between 1997 and 2007 led to a -5.5% decrease in manufacturing markups, and a -0.4% decrease in non-manufacturing markups. The surge of imports also leads to a 6.1% increase in real consumption, 20% more than with fixed markups. In this context, imports reduced net markups, and the gains from trade are larger the more variable markups are. This study serves as a first approximation to general equilibrium models featuring both competitive effects of imports and their link to the gains from trade.

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1 Introduction

Imports to the U.S. rose from 10.1% of GDP in 1991 to an all time high of 17.4% in 2008¹. Imports are beneficial because they provide access to cheaper goods for consumers and firms. But imports are also be beneficial because they reduce markups, increasing efficiency as domestic producers compete with foreign providers of similar products. This is called the pro-competitive effect of trade, and it is both suggested by theory and confirmed empirically.

However, just as there are losers in specialization, imports also have a downside for competition. Recent studies have found that imports can increase the markups of buying firms, in particular when the surge is in foreign products used as inputs in production. This has been called the anti-competitive effect of imports, and discussed for example in [Martynov and Zhang \[2023\]](#), [Impullitti and Kazmi \[2023\]](#), [De Loecker et al. \[2016\]](#), [Amiti and Konings \[2007\]](#). The existence of both types of competitive effects means that imports can both decrease and increase domestic markups.

Because of this ambiguity, we need more structure to disentangle the net effect of imports on welfare. To that end, I present a model incorporating both pro-competitive and anti-competitive effects of trade, to quantitatively evaluate the role of markup adjustment. The model features both effects in a multi-sector small open economy with inputs in production, trade, and input-output linkages. Variable markups and incomplete cost pass-through result from assuming variable elasticities of demand. Then, when imports compete with domestic goods, they lead to lower domestic markups; and when imports of inputs decrease costs, they lead to higher domestic markups.

Through my model, I quantify the change in markups resulting from the surge of imports to the U.S. between 1997 and 2007. In particular, I match the shares of domestic expenditure for each sector and use, constructed from U.S. national accounts data. I find the net competitive effect of imports is to reduce manufacturing markups by -5.5% , and non-manufacturing markups by -0.4% . The surge of imports also leads to a 6.1% increase

¹U.S. Bureau of Economic Analysis

in real consumption, 20% more than with fixed markups. In this context, imports reduced markups and increased the gains from trade.

I expand on previous analysis by [Arkolakis et al. \[2019\]](#), focusing on the domestic economy. In their approach, the effect of trade on markups is ambiguous *across countries*: foreign markups increase due to lower trade costs, and domestic markups decrease due to import competition. In my model, the effect of trade on markups is ambiguous *within the country*: some domestic markups increase through lower input costs, as other domestic markups decrease through competition. In addition, I allow for sectors to reallocate in response to both imports and markups, in both quantity produced and factors used for production.

The anti-competitive effect requires accounting for the interaction of trade with the input-output organization of production. By input-output structure I mean a framework where firms use the output of other firms as an input for production; in turn selling part of their production as inputs for other firms. [Baqae and Farhi \[2023\]](#) show how the input-output structure and market distortions affect the gains from trade, but they use fixed distortions in their approximation. The production loop can also amplify even modest changes in markups, as shown by [Bridgman and Herrendorf \[2023\]](#).

My work uses Kimball demand in a model with trade in inputs and final goods, borrowing from [Comin and Johnson \[2022\]](#). I extend this framework to multiple sectors with input-output linkages, necessary to distinguish anti-competitive effects from pro-competitive effects. On the other hand, I drop sticky prices and other features of the Neo-Keynesian model, as I focus on markup levels and not inflation. My model is also related to [Gopinath and Itskhoki \[2010\]](#) and [Gopinath et al. \[2020\]](#) given the variable markup structure and role of pass-through on both, but I focus on cost pass-through and not exchange-rate pass-through.

In Section 2 I present a model to quantify the competitive effect of imports and their welfare contribution. Section 3 highlights the mechanism of competitive effects and discusses the intuition on how markups affect the gains from trade. Section 4 presents the results of computing my model. Section 5 concludes.

2 Model

This section presents the model, a small open economy multi-sector framework with variable markups. It features inputs in production, multiple sectors intertwined in an input-output structure, and trade in both intermediate and final goods. Variable markups arise from using Kimball technology aggregation and preferences over varieties. The model is static, and prices are flexible. There will be two types agents: consumers and firms.

2.1 Consumer

Starting with the consumer side, I will assume that preferences for the representative consumer over consumption C_t and labor supply L_t are:

$$U(C_t, L_t) = \frac{C_t^{1-\rho}}{1-\rho} - \varsigma \frac{L_t^{\psi+1}}{\psi+1}, \quad (1)$$

where ρ governs the marginal utility of consumption, ψ is the inverse of the Frisch elasticity of labor supply, and ς is a scaling parameter.

Aggregate consumption C_t is a basket of final goods C_{st} from each sector $s \in \{1, \dots, S\}$, with CES preferences over the sector-composite goods:

$$C_t = \left(\sum_s \zeta_s^{\frac{1}{\vartheta}} C_{st}^{\frac{\vartheta-1}{\vartheta}} \right)^{\frac{\vartheta}{\vartheta-1}}, \quad (2)$$

where ζ_s controls the consumer preference for the sector s composite good, and ϑ is the elasticity of substitution for consumption across sectors.

Within each sector, there is a unit continuum of Home varieties and a unit continuum of Foreign varieties. Consumer preferences over Home and Foreign varieties are given by:

$$\nu_s \int_0^1 \Upsilon \left(\frac{C_{Hist}}{\nu_s C_{st}} \right) di + (1 - \nu_s) \int_0^1 \Upsilon \left(\frac{C_{Fist}}{(1 - \nu_s) C_{st}} \right) di = 1, \quad (3)$$

where function $\Upsilon(\cdot)$ must satisfy $\Upsilon(1) = 1$, $\Upsilon'(\cdot) > 0$, and $\Upsilon''(\cdot) < 0$, and ν_s is the home bias parameter. This form follows the definition by [Kimball \[1995\]](#). Furthermore, I adopt the definition of $\Upsilon(\cdot)$ proposed by [Klenow and Willis \[2016\]](#):

$$\Upsilon(x_s) = 1 + (\sigma_s - 1) \exp\left(\frac{1}{\epsilon_s}\right) \epsilon_s^{\frac{\sigma_s}{\epsilon_s - 1}} \left(\Gamma\left(\frac{\sigma_s}{\epsilon_s}, \frac{1}{\epsilon_s}\right) - \Gamma\left(\frac{\sigma_s}{\epsilon_s}, \frac{x_s^{\frac{\epsilon_s}{\sigma_s}}}{\epsilon_s}\right) \right), \quad (4)$$

where $\Gamma(u, z) = \int_z^\infty s^{u-1} e^{-s} ds$ is the incomplete gamma function, and $\Upsilon(\cdot)$ has sector-specific parameters σ_s and ϵ_s , with $\sigma_s > 1$ and $\epsilon_s > 0$. In particular, σ_s determines the steady state symmetric elasticity of demand, and through it also markups, while ϵ_s controls the variability of markups.

Expenditure will then be the sum of expenditures across sectors:

$$P_{Ct} C_t = \sum_s P_{Cst} C_{st}, \quad (5)$$

with P_{Cst} being the composite price of consumption from sector s , which combines Home and Foreign varieties:

$$P_{Cst} C_{st} = \int_0^1 P_{Hist} C_{Hist} di + \int_0^1 P_{Fist} C_{Fist} di, \quad (6)$$

where P_{Hist} and P_{Fist} are the prices for each individual variety, produced by Home or Foreign economies.

Consumers finance expenditure through wages W_t earned from supplying labor L_t to firms and by collecting profits Π_t from the domestic firms they own. They also receive an exogenous transfer T_t from the Foreign country (the rest of the world), which allows for trade imbalances in this static model. The nominal flow budget constraint is then:

$$P_{Ct} C_t = W_t L_t + \Pi_t + T_t, \quad (7)$$

where P_{Ct} is the price of aggregate consumption.

Consumer Problem The consumer solves the following problem. Given prices $\{P_{Ct}, P_{Cst}, P_{Hist}, P_{Fist}, W_t\}$ and foreign transfer $\{T_t\}$, the consumer chooses consumption $\{C_t, C_{st}, C_{Hist}, C_{Fist}\}$ and labor supply $\{L_t\}$ to maximize utility (equation (1) with preferences defined in equations (2-6)) subject to its budget constraint (equation (7)).

The optimal decision for consumption and labor is thus determined by consumption prices P_{Ct} and wages W_t :

$$C_t^{-\rho} \frac{W_t}{P_{Ct}} = \varsigma L_t^\psi, \quad (8)$$

where the price of the aggregate consumption good is a CES composite of prices for the individual sector-level composite goods and parameters ζ_s and ϑ :

$$P_{Ct} = \left(\sum_s \zeta_s P_{Cst}^{1-\vartheta} \right)^{\frac{1}{1-\vartheta}}. \quad (9)$$

The sector composition of the consumption basket C_{st} depends on the ratio of prices $\frac{P_{Cst}}{P_{Ct}}$, and on parameters ζ_s and ϑ :

$$C_{st} = \zeta_s \left(\frac{P_{Cst}}{P_{Ct}} \right)^{-\vartheta} C_t / \quad (10)$$

The corresponding price of consumption from each sector is a weighted average of prices from each source:

$$P_{Cst} = P_{Hist} \frac{C_{Hist}}{C_{st}} + P_{CFst} \frac{C_{Fst}}{C_{st}}.$$

Optimal demand for the consumption goods of each source, Home and Foreign, is:

$$C_{Hist} = \nu_s \Psi \left(D_{Cst} \frac{P_{Hist}}{P_{Cst}} \right) C_{st}, \quad (11)$$

$$C_{Fist} = (1 - \nu_s) \Psi \left(D_{Cst} \frac{P_{Fist}}{P_{Cst}} \right) C_{st}, \quad (12)$$

where D_{Cst} is a demand index factor:

$$D_{Cst} = \int_0^1 \Upsilon' \left(\frac{C_{Hist}}{\nu_s C_{st}} \right) \frac{C_{Hist}}{C_{st}} di + \int_0^1 \Upsilon' \left(\frac{C_{Fist}}{(1 - \nu_s) C_{st}} \right) \frac{C_{Fist}}{C_{st}}, \quad (13)$$

with prices for domestic varieties P_{Hist} determined by each firm. Function $\Psi(.) = \Upsilon'^{-1}(.)$ follows from the [Klenow and Willis \[2016\]](#) definition of $\Upsilon(.)$:

$$\Psi(y) = \Upsilon'^{-1}(y) = \left(1 + \epsilon_s \ln \left(\frac{\sigma_s - 1}{\sigma_s y} \right) \right)^{\frac{\sigma_s}{\epsilon_s}}. \quad (14)$$

2.2 Firms

Each firm produces a single differentiated variety i in sector s , combining labor and inputs to produce gross output quantity Y_{ist} . The production technology for each variety is Cobb-Douglas:

$$Y_{ist} = Z_{st} L_{ist}^{1-\alpha_s} M_{ist}^{\alpha_s}, \quad (15)$$

where L_{ist} is labor, and M_{ist} is a composite of input varieties across sectors from Home and Foreign. Here, total factor productivity Z_{st} and the production elasticity α_s are sector specific.

The composite input used by firm i in sector s is a CES combination of inputs from each

supplying sector s' :

$$M_{ist} = \left(\sum_{s'} \left(\frac{\alpha_{s's}}{\alpha_s} \right)^{\frac{1}{\kappa_s}} M_{s'ist}^{\frac{\kappa_s-1}{\kappa_s}} \right)^{\frac{\kappa_s}{\kappa_s-1}}, \quad (16)$$

with κ_s representing the substitution of inputs across sectors. $\alpha_{s's}$ controls demand by sector s for inputs from sector s' , where $\sum_{s'} \alpha_{s's} = \alpha_s$ ².

In similar fashion as for consumption, I assume varieties from Home and Foreign economies are aggregated using Kimball technology, implicitly defined as:

$$\xi_{s's} \int_0^1 \Upsilon \left(\frac{M_{Hjs'ist}}{\xi_{s's} M_{s'ist}} \right) dj + (1 - \xi_{s's}) \int_0^1 \Upsilon \left(\frac{M_{Fjs'ist}}{(1 - \xi_{s's}) M_{s'ist}} \right) dj = 1, \quad (17)$$

where $\Upsilon(\cdot)$ is defined in the same manner as in consumption. $\xi_{s's}$ is the home bias of firms from sector s when buying inputs from sector s' . $M_{Hjs'ist}$ and $M_{Fjs'ist}$ are quantities used for domestic production, where variety j of sector s' is sold to firm i in sector s , produced at either Home and Foreign. $M_{s'ist}$ is then the quantity of inputs sold by sector s' to firms i in sector s .

Each firm can sell its output on the domestic market Y_{Hist} or as exports X_{ist} , such that $Y_{ist} = Y_{Hist} + X_{ist}$. Firms can also set different prices for exports P_{Xist} and domestic sales P_{Hist} , but not across destinations at Home. Profits for each firm in a sector are then the revenue from both from domestic sales and exports, minus the production costs of labor and inputs:

$$\Pi_{ist} = P_{Hist} Y_{Hist} + P_{Xist} X_{ist} - C(W_t, P_{Mist}), \quad (18)$$

where I assume there are no fixed costs, so the cost function $C(\cdot)$ is defined as:

$$C(W_t, P_{Mist}) = W_t L_{ist} + P_{Mist} M_{ist}. \quad (19)$$

where I assume individual firms do not price-discriminate between their domestic buyers, be it consumers or firms.

²Note that if $\kappa = 1$, we would have $M_{ist} = \prod_{s'} (M_{s'ist})^{\frac{\alpha_{s's}}{\alpha_s}}$ which makes $Y_{ist} = Z_{st} L_{ist}^{1-\alpha_s} \prod_{s'} (M_{s'ist})^{\alpha_{s's}}$.

Firm Problem For given prices $\{P_{Mist}, P_{Hi'st}, P_{Fist}, P_{Hjs'ist}\}$, each firm maximizes profits (18) by choosing its' domestic production Y_{Hist} and prices P_{Hist} , export production X_{ist} and prices $P_{X'ist}$, use of labor $\{L_{ist}\}$ and inputs $\{M_{ist}, M_{Hist}, M_{Fist}, M_{Hjs'ist}\}$; subject to the cost function (19) and technologies (15-17). Marginal cost $\{MC_{it}\}$ is then a byproduct of optimal cost.

Optimal demands for labor L_{ist} and total inputs M_{ist} are conditional on firm output Y_{ist} , the marginal costs of each variety MC_{ist} and the elasticities of production α_s , given wages W_t and input prices P_{Mist} :

$$W_t L_{ist} = (1 - \alpha_s) Y_{ist} MC_{ist} \quad (20)$$

$$P_{Mist} M_{ist} = \alpha_s Y_{ist} MC_{ist}. \quad (21)$$

The marginal cost for firms in sector s is also a function of wages W_t and the price of the input basket P_{Mist} , as well as sector productivity Z_{st} and production elasticity α_s :

$$MC_{ist} = Z_{st}^{-1} (1 - \alpha_s)^{-(1-\alpha_s)} \alpha_s^{-\alpha_s} W_t^{(1-\alpha_s)} P_{Mist}^{\alpha_s}. \quad (22)$$

$M_{s'is}$ is the demand of each variety i in sector s for inputs from each sector s' ; it is determined by the total demand for inputs in sector s , M_{ist} , the production parameters $\frac{\alpha_{s's}}{\alpha_s}$, the price ratio for each input with respect to the basket $\frac{P_{Ms'ist}}{P_{Mist}}$, and the elasticity of substitution κ_s :

$$M_{s'ist} = M_{ist} \left(\frac{\alpha_{s's}}{\alpha_s} \right) \left(\frac{P_{Ms'ist}}{P_{Mist}} \right)^{-\kappa_s}, \quad (23)$$

with corresponding CES prices for the input basket:

$$P_{Mist} = \left(\sum_{s'} \left(\frac{\alpha_{s's}}{\alpha_s} \right) P_{Ms'ist}^{1-\kappa_s} \right)^{\frac{1}{1-\kappa_s}}. \quad (24)$$

As in consumption, the optimal demand for inputs from each source is determined by the ratio

of prices $\frac{P_{His't}}{P_{Ms'ist}}$ times the demand factor $D_{Ms'st}$ as arguments of function $\Psi(\cdot)$:

$$M_{His'ist} = \xi_{s's} \Psi \left(D_{Ms'ist} \frac{P_{His't}}{P_{Ms'ist}} \right) M_{s'ist} \quad (25)$$

$$M_{Fs'ist} = (1 - \xi_{s's}) \Psi \left(D_{Ms'ist} \frac{P_{MFs't}}{P_{Ms'ist}} \right) M_{s'ist}, \quad (26)$$

where $\Psi(\cdot)$ is as defined previously in the consumer problem, with the same parameters. The price of inputs from each sector s' to firm i in sector s is the weighted average across sources:

$$P_{Ms'ist} = \int_0^1 P_{MHs'ist} \frac{M_{His'ist}}{M_{s'ist}} di + \int_0^1 P_{MFs'ist} \frac{M_{Fs'ist}}{M_{s'ist}} di, \quad (27)$$

where P_{Hist} is the price of each firm.

The optimal pricing for each individual firm is a markup over marginal costs:

$$P_{Hist} = \frac{\epsilon_{Hist}}{\epsilon_{Hist} - 1} MC_{ist}, \quad (28)$$

with ϵ_{Hist} being the variable elasticity of demand faced by monopolistic firms i in sector s , which is itself a weighted average of the variable elasticities of demand from each market the firm serves:

$$\epsilon_{Hist} = \epsilon_{Hist}^C \frac{C_{Hist}}{C_{Hist} + \sum_{s'} M_{His'st}} + \sum_{s'} \epsilon_{His'st}^M \frac{M_{His'st}}{C_{Hist} + \sum_{s'} M_{His'st}} \quad (29)$$

Preferences for exports are CES with elasticity ϵ_{Xs} , making the optimal price of exports a fixed markup over marginal costs:

$$P_{Xist} = \frac{\epsilon_{Xs}}{\epsilon_{Xs} - 1} MC_{ist}, \quad (30)$$

2.3 Imports, Market Clearing, and Export Demand

Each Foreign firm also produces a single differentiated variety i in sector s . I assume foreign marginal costs are given, and no fixed costs, so the profits for each firm in a sector are the difference

between prices and marginal costs times gross output:

$$\Pi_{ist}^* = (P_{Fist}^* - MC_{Fist}^*) Y_{ist}^*, \quad (31)$$

where the firm maximizes profits (31) for given marginal costs MC_{ist}^* , and there is no price discrimination across Home buyers.

The optimal pricing of imports for each Foreign firm is again markup over marginal costs:

$$P_{Fist}^* = \frac{\epsilon_{Fist}}{\epsilon_{Fist} - 1} MC_{ist}^*, \quad (32)$$

with ϵ_{Fist} being the variable elasticity of demand for foreign goods bought by domestic buyers of the monopolistic Foreign firms i in sector s . This elasticity is once again a weighted average of the variable elasticities of demand from each market the firm serves:

$$\epsilon_{Fist} = \epsilon_{Fist}^C \frac{C_{Fist}}{C_{Fist} + \sum_{s'} M_{Fis'st}} + \sum_{s'} \epsilon_{Fis'st}^M \frac{M_{Fis'st}}{C_{Fist} + \sum_{s'} M_{Fis'st}}. \quad (33)$$

I allow for imports of consumption and input goods to face different trade frictions. I combine these frictions with the exogenous foreign marginal costs, so the optimal price of imports is again markups over these (bundled) marginal costs:

$$P_{CFst} = \frac{\epsilon_{Fst}}{\epsilon_{Fst} - 1} MC_{Cst}, \quad (34)$$

$$P_{MFs'st} = \frac{\epsilon_{Fst}}{\epsilon_{Fst} - 1} MC_{Ms'st}. \quad (35)$$

Note that this means imports also have variable markups and incomplete pass-through in my model.

Market clearing at Home for each variety i in sector s is achieved when its production equals the sum across all its uses:

$$Y_{ist} = C_{Hist} + \int_0^1 M_{Hij's't} dj + X_{ist}, \quad (36)$$

Likewise, the clearing condition for the labor market requires labor supply to equal the amount

of labor used to make each variety i in each sector s :

$$L_t = \sum_s \int_0^1 L_{ist} di. \quad (37)$$

Foreign demand for exports is driven by exogenous foreign consumption C_{st}^* , with preferences over exports and foreign production following a CES function:

$$C_{st}^* = \left((C_{fst}^*)^{\frac{\eta_{Xs}-1}{\eta_{Xs}}} + (X_{st})^{\frac{\eta_{Xs}-1}{\eta_{Xs}}} \right)^{\frac{\eta_{Xs}}{\eta_{Xs}-1}}, \quad (38)$$

where η_{Xs} is the elasticity substitution between sources. Preferences over varieties of exports also follow a CES structure such that:

$$X_{st} = \left(\int_0^1 (X_{ist}^*)^{\frac{\varepsilon_{Xs}-1}{\varepsilon_{Xs}}} di \right)^{\frac{\varepsilon_{Xs}}{\varepsilon_{Xs}-1}}, \quad (39)$$

where ε_{Xs} is the elasticity of substitution across varieties of exports.

Optimal exports for each variety X_{ist} are then determined as:

$$X_{ist} = \left(\frac{P_{Xist}}{P_{Xst}} \right)^{-\varepsilon_{Xs}} X_{st}, \quad (40)$$

with P_{Xist} the variety price of exports set by domestic firms and P_{Xst} the price of the sector composite of exports. The optimal quantity of sector composite exports is set as:

$$X_{st} = \left(\frac{P_{Xst}}{P_{Cst}^*} \right)^{-\eta_{Xs}} C_{st}^*, \quad (41)$$

where P_{Cst}^* is the exogenous price of foreign consumption, and C_{st}^* again the exogenous foreign demand.

2.4 Equilibrium

I will focus on the equilibrium where firms i are symmetric, collapsing to a representative firm per sector s , and therefore dropping the subscript i . Given the values for $\{P_{CFst}, P_{MFs'st}, Z_{st}, P_{Cst}^*,$

$C_{st}^*, T_t\}$, the equilibrium is a group of 8 sets of prices $\{P_{Hst}, MC_{st}, P_{Ct}, P_{Cst}, P_{Mst}, P_{Ms'st}, P_{Xst}, W_t\}$ and 15 sets of allocations $\{L_t, L_{st}, C_t, C_{st}, C_{Hst}, C_{Fst}, D_{Cst}, M_{st}, M_{s'st}, M_{Hs'st}, M_{Fs'st}, D_{Ms'st}, X_{st}, Y_{st}, \Pi_t\}$ that maximize utility for the consumer, maximize profits for firms, and clear markets for goods and labor. The full set of equilibrium conditions can be found in Appendix 1.

3 Discussion

In this section, I present a brief discussion on how the competitive effects of trade operate to increase welfare. I begin by highlighting the two ways in which import competition affects markups in my model. I follow by presenting some intuition on how markups combine to impact the gains from trade.

3.1 Import Competition and Markups

In this model, the competitive effects of imports will work by either adding competitive pressure in sales, or diminishing pressure through costs, yielding the pro-competitive and anti-competitive mechanisms. I describe how each mechanism works using Kimball preferences and technology.

3.1.1 Pro-Competitive Effect

The pro-competitive effect can be thought of as the optimal response of imperfectly competitive firms to an increase in imports. Essentially, when imports enter a specific market, they reduce prices in the sector to the benefit of consumers and detriment of domestic firms. This price reduction can be decomposed into two channels. The more straightforward channel is the plain substitution of sales, whereby foreign production replaces domestic supply at a lower price. But incumbent firms also react to the entry of imports. Facing tougher competition, and assuming they have positive margins, domestic suppliers can lower their own prices to avoid losing too many sales. Thus, imports are beneficial not only because they decrease prices but also because the increase in competition lowers domestic markup distortions.

Kimball preferences deliver the pro-competitive effect of trade as follows. The optimal markup for firm i will vary with the elasticity of demand, which in turn is determined by the ratio between the firm's price P_{Hst} and the aggregate price level in the market it serves, either P_{Cst} in consumption goods or $P_{Ms'st}$ in inputs. In the symmetric equilibrium, the optimal markups μ_{Hst} are:

$$\mu_{Hst} = \frac{\epsilon_{Hst}}{\epsilon_{Hst} - 1}, \quad (42)$$

where the elasticity of demand for the goods of sector s is a weighted average of the elasticities in each market:

$$\varepsilon_{Hst} = \varepsilon_{Hst}^C \frac{C_{Hist}}{C_{Hist} + \sum_{s'} M_{Hss't}} + \sum_{s'} \varepsilon_{Hss't}^M \frac{M_{Hss't}}{C_{Hist} + \sum_{s'} M_{Hss't}}, \quad (43)$$

and the elasticity of demand in each market is:

$$\varepsilon_{Hst}^C = \sigma_s \left(1 + \epsilon_s \ln \left(\frac{\sigma_s}{\sigma_s - 1} \right) - \epsilon_s \ln \left(D_{Cst} \frac{P_{Hst}}{P_{Cst}} \right) \right)^{-1}, \quad (44)$$

$$\varepsilon_{Hss't}^M = \sigma_s \left(1 + \epsilon_s \ln \left(\frac{\sigma_s}{\sigma_s - 1} \right) - \epsilon_s \ln \left(D_{Mss't} \frac{P_{Hst}}{P_{Mss't}} \right) \right)^{-1}. \quad (45)$$

This describes the nature of variable markups. Setting demand indices D_{Cst} and $D_{Mss't}$ aside, and even though domestic firms are symmetric, markups are variable insofar as the relation between domestic prices and aggregate prices $\frac{P_{Hst}}{P_{Cst}}$ and $\frac{P_{Hst}}{P_{Mss't}}$ is affected by foreign prices P_{CFst} and $P_{MFs't}$ through the respective denominators. For instance, when there is a surge in imports of final goods, in this model, P_{CFst} decreases, driving consumer prices P_{Cst} downward. The domestic firm in turn will lower its markups μ_{Hst} as its elasticity of demand ε_{Hst}^C increases.

3.1.2 Anti-Competitive Effect

The anti-competitive effect is also just the optimal response of imperfectly competing firms to an increase in imports, but now with respect to the import of inputs. For example, when imports lower prices in one market, downstream firms can buy their inputs for less, and thereby lower their costs of production. With imperfect competition, firms will not fully transmit their cost reduction to their customers: they will increase their markups while reducing prices.

The anti-competitive effect operates through Kimball preferences, as they also deliver variable cost pass-through. More specifically, the assumed preferences not only provide variable elasticities of demand but also a variable rate of change for those elasticities, or a variable super-elasticity of demand. It is this super-elasticity that determines the cost pass-through, and an incomplete cost pass-through that creates the anti-competitive effect.

Under the formulation of Kimball preferences used in my model, the super-elasticity of demand

is defined, first for consumption goods, as:

$$\Gamma_{Hst}^C = \frac{\epsilon_s}{\left(\sigma_s - 1 - \epsilon_s \ln\left(\frac{\sigma_s - 1}{\sigma_s}\right) + \epsilon_s \ln\left(D_{Cst} \frac{P_{Hst}}{P_{Cst}}\right)\right)}, \quad (46)$$

which makes the firm's cost pass-through Φ_{Hst}^C :

$$\Phi_{Hst}^C = \frac{1}{1 + \Gamma_{Hst}^C}. \quad (47)$$

Likewise for inputs, I will have input super-elasticity:

$$\Gamma_{Hs'st}^M = \frac{\epsilon_s}{\left(\sigma_s - 1 - \epsilon_s \ln\left(\frac{\sigma_s - 1}{\sigma_s}\right) + \epsilon_s \ln\left(D_{Ms'st} \frac{P_{Hst}}{P_{Ms't}}\right)\right)}, \quad (48)$$

as well as input cost pass-through $\Phi_{Hs'st}^M$:

$$\Phi_{Hst}^C = \frac{1}{1 + \Gamma_{Hs'st}^M}. \quad (49)$$

In the CES case, the elasticities are fixed so that the super-elasticity $\Gamma_{Ht} = 0$ and the cost pass-through $\Phi_{Ht} = 1$. But in the Kimball case, both markups and pass-through change with the price ratio. Note also that the imperfect pass-through operates both through foreign inputs $P_{MFs't}M_{Fs'st}$ and domestic inputs $P_{Hs't}M_{Hs'st}$ supplied by other sectors.

To illustrate how incomplete cost pass-through matters, consider the response of a domestic firm when the foreign price of an input changes. Assume the price of a foreign input drops, and the resulting re-optimization yields a drop in marginal costs of 10%. In the CES case, demand elasticities are fixed and pass-through is complete, so prices drop by the same 10%. However, with Kimball preferences, the same 10% creates a smaller price drop, say 5%, as the demand elasticity faced by the domestic firm is reduced. At the same time, the reduction in demand elasticity means that markups adjust up. The anti-competitive effect is thus a consequence of the cost-price pass-through Φ_{Ht}^C being less than one.

3.2 Markups and the Gains from Trade

In this section, I present the intuition on how markups link to the gains from trade in my model. I first present the main intuition in a one-sector example and then extend it to multiple sectors.

3.2.1 One-Sector Example

The gains from trade are the improvement in welfare countries reap from engaging in international trade. To analyze it in a simple environment, take a one-sector version of the model presented earlier and assume balanced trade $T_t = 0$ for clarity.

Gains from Trade with One Sector In the presented setting, consumption is financed by wages and profits according to the usual budget constraint:

$$P_{Ct}C_t = W_tL_t + \Pi_t, \quad (50)$$

where C_t is real consumption, sold at price P_{Ct} , W_t are the wages paid to labor L_t , and Π_t are firm profits.

The wage bill is tied to production through labor demand, in this case:

$$W_tL_t = (1 - \alpha) (MC_t) Y_t, \quad (51)$$

with output being produced as a Cobb-Douglas combination of labor and inputs $Y = ZL^{(1-\alpha)}M^\alpha$, and MC_t representing marginal costs. Firms will sell their production at Home or to the Foreign economy. As optimal pricing corresponds to markups times marginal costs, profits will take the form:

$$\Pi_t = \frac{1}{\epsilon_{Ht}} P_{Ht} Y_{Ht} + \frac{1}{\epsilon_{Xt}} P_{Xt} X_t, \quad (52)$$

where Y_{Ht} is the part of output sold within the Home economy at price P_H , facing elasticity ϵ_H . Likewise, Y_X is the part of output sold as exports at price P_X , facing elasticity ϵ_X . I can rewrite

this expression as:

$$\Pi_t = \frac{1}{\epsilon_{Ht}} P_{Ht} (Y_{Ht} + X_t) + \left(\frac{1}{\epsilon_{Xt}} P_{Xt} - \frac{1}{\epsilon_{Ht}} P_{Ht} \right) X_t, \quad (53)$$

where the first term represents total production sold at Home prices, and the second term reflects the additional profits from pricing-to-market of exports.

Combining equations (50-51) and (53) gives:

$$P_{Ct} C_t = (1 - \alpha) (MC_t) Y_t + \frac{1}{\epsilon_{Ht}} P_{Ht} (Y_{Ht} + X_t) + \left(\frac{1}{\epsilon_{Xt}} P_{Xt} - \frac{1}{\epsilon_{Ht}} P_{Ht} \right) X_t. \quad (54)$$

Using optimal pricing again, and expressing markups as functions of the demand elasticity $\mu_{Ht} = \frac{\epsilon_{Ht}}{\epsilon_{Ht}-1}$, I obtain an expression for real consumption as a function of markups:

$$C_t = \left(1 - \frac{\alpha}{\mu_{Ht}} \right) \frac{P_{Ht}}{P_{Ct}} Y_t + \left(\frac{1}{\epsilon_{Xt}} P_{Xt} - \frac{1}{\epsilon_{Ht}} P_{Ht} \right) \frac{X_t}{P_{Ct}}. \quad (55)$$

Equation (55) contains many familiar components from the gains from trade: First, the terms of trade $\frac{P_{Ht}}{P_{Ct}}$ represent how expensive domestic production is with respect to consumption. This can be thought of as the classical gains from trade, where a surge in imports lowers P_C for a given P_H , increasing welfare.

Second, import competition changes production Y_t through three mechanisms: (a) changes in marginal costs due to input prices, which I call cost gains from trade; (b) reallocation across factors of production M_t and L_t ; and (c) changes in allocative efficiency due to input markups μ_{Ht} . I explore these channels in more depth in the following section.

Third, imports change the term $\left(1 - \frac{\alpha}{\mu_{Ht}} \right)$ through markups μ_{Ht} , which affects the share of output that turns into domestic resources. For example, if only labor was used in production, $\alpha = 0$, all the value of production would be available to finance consumption.

Fourth, there is a profit-shifting term, given by the price difference between exports and domestic prices. I will mostly abstract from this mechanism, both for simplicity and because it cancels out in my calibration of the symmetric equilibrium.

Here, markups will play two roles. First, lower markups decrease the term that drives how much

of production turns into consumption $\left(1 - \frac{\alpha}{\mu_H}\right)$, as lower markups mean lower profits $\Pi = \left(1 - \frac{1}{\mu_H}\right) Y$. At the same time, lower markups indirectly help increase output because domestic input prices are lower.

Consumption and Markups To unpack the effect of markups in production, I use the first order approximations of the model presented in section 3. I also abstract from the pricing-to-market term to focus on import competition and markups. Starting with equation (55), I have:

$$\hat{c}_t = \frac{\alpha}{\mu_{H0} - \alpha} \hat{\mu}_{Ht} + \hat{p}_{Ht} - \hat{p}_{Ct} + \hat{y}_t, \quad (56)$$

where terms $\hat{x}$ are the log deviations from baseline. As presented earlier, we will have the first term directly linked to markups, a second term related to the terms of trade, and a last term encompassing the effects on production. However, in equation (56), changes in markups work in the same direction as changes in real consumption, counterintuitively suggesting that higher markups increase welfare. This is because up to this point, I only consider the effect of markups on profits for a given level of production. It will be the effect of markups on production that will revert this sign.

To disentangle the term linked to production, assume for now that both productivity and labor are fixed. Then, changes in production would be ruled by changes in input use:

$$\hat{y}_t = \alpha \hat{m}_t, \quad (57)$$

where $\hat{m}_t$ is now the change in input use. Log-linearizing the optimal demand for inputs $P_{Mt} M_t = \alpha (MC_t) Y_t$, in combination with the optimal pricing expressed as $MC_t = \frac{P_H}{\mu_H}$, I can write the change in inputs as

$$\hat{m}_t = (p_{Ht} - p_{Mt}) + y_t - \mu_{Ht}, \quad (58)$$

that is, the change in inputs is a function of input prices, markups, and production. Now, combine

(57) and (58) to obtain an expression for changes in production:

$$\hat{y} = \frac{\alpha}{1-\alpha} (p_H - p_M) - \frac{\alpha}{1-\alpha} \mu_H, \quad (59)$$

which helps track down the response of production to an increase in imports. The first term suggests that, for given domestic prices, a decrease in the price of the input basket will induce higher production. These are the cost gains from trade at work, with imports reducing the cost of inputs and thereby increasing efficiency. The second term represents the inverse relation between markup distortion and output.

Focusing back on welfare, I can combine equations (56) and (59) to deliver a formula for the evolution of real consumption in terms of markups and prices:

$$\hat{c}_t = \frac{\alpha}{\mu_{H0} - \alpha} \hat{\mu}_{Ht} + (\hat{p}_{Ht} - \hat{p}_{Ct}) + \frac{\alpha}{1-\alpha} (\hat{p}_{Ht} - \hat{p}_{Mt}) - \frac{\alpha}{1-\alpha} \mu_{Ht}. \quad (60)$$

Equation (60) disentangles the effects on welfare. The first term $\frac{\alpha}{\mu_{H0} - \alpha} \hat{\mu}_{Ht}$ gives us the increase in resources for consumers from larger profits. The second term $(\hat{p}_{Ht} - \hat{p}_{Ct})$ gives the classical terms-of-trade gains. The third term $\frac{\alpha}{1-\alpha} (p_H - p_M)$ gives the cost-channel gains from less expensive inputs. The final term $-\frac{\alpha}{1-\alpha} \mu_H$ gives us the allocative efficiency change due to changes in the markup distortion. Rearranging the previous expression also provides some intuition:

$$\hat{c}_t = \left(\frac{\alpha}{\mu_{H0} - \alpha} - \frac{\alpha}{1-\alpha} \right) \hat{\mu}_{Ht} + (\hat{p}_{Ht} - \hat{p}_{Ct}) + \frac{\alpha}{1-\alpha} (\hat{p}_{Ht} - \hat{p}_{Mt}). \quad (61)$$

Equation (61) shows how a decrease in markups will always increase welfare. This is because for any change in markups, the loss in resources from markups will always be smaller than the resources gained from improving efficiency. This will always be case as long as markups at baseline are above one $\mu_{H0} > 1$, making $\frac{\alpha}{\mu_{H0} - \alpha} < \frac{\alpha}{1-\alpha}$

Domestic Expenditure Shares Equation (61) can also be rewritten in terms of the domestic expenditure shares, which [Arkolakis et al. \[2012\]](#) identify as sufficient statistics for welfare. In

particular, the change in prices can be rewritten as:

$$(\hat{p}_{Ht} - \hat{p}_{Ct}) = -\frac{1}{\sigma - 1} \lambda^C \quad (62)$$

$$(p_H - p_M) = -\frac{1}{\sigma - 1} \lambda^M. \quad (63)$$

Likewise, the change in markups will be linked to the change in demand elasticities, which in turn are linked to expenditure shares such that:

$$\hat{\mu}_{Ht} = \mu_{H0} \frac{1}{\sigma - 1} \left(\frac{C_{H0}}{Y_{H0}} \hat{\lambda}_{Ht}^C + \frac{M_{H0}}{Y_{H0}} \hat{\lambda}_{Ht}^M \right), \quad (64)$$

in other words, larger expenditure shares increase markups, diminishing real consumption. Combining equations (61-64) yields a formula for real consumption as a function of domestic expenditure shares:

$$\hat{c}_t = -\frac{1}{\sigma - 1} \left[J_1 \mu_{H0} \left(\frac{C_{H0}}{Y_{H0}} \hat{\lambda}_{Ht}^C + \frac{M_{H0}}{Y_{H0}} \hat{\lambda}_{Ht}^M \right) + \hat{\lambda}_{Ht}^C + \frac{\alpha}{1 - \alpha} \hat{\lambda}_{Ht}^M \right], \quad (65)$$

with $J_1 = \left(\frac{\alpha}{1 - \alpha} - \frac{\alpha}{\mu_{H0} - \alpha} \right) > 0$.

Equation (65) shows how the domestic expenditure shares are again a sufficient statistic of welfare. But even more important, it shows how the gains from trade are larger with variable markups. More specifically, with fixed markups the first term inside the brackets is null, as imports have no effect on domestic markups. But when I allow imports to have competitive effects, lower markups contribute to welfare by inducing higher efficiency.

3.2.2 Multi-Sector Example

One advantage of the one-sector example is that the price of inputs is the same as the price paid for domestic inputs. However, and for the same reason, the markup-increasing and markup-decreasing effects will be superimposed, as there is only one markup experiencing both the pro-competitive and anti-competitive effects of imports.

To circumvent this limitation, I derive a multi-sector version of equation (61), following a similar

procedure. Starting with the multi-sector version of equation (56):

$$\hat{c}_t = \sum_s \kappa_s \left(\frac{\alpha_s}{\mu_{Hs0} - \alpha_s} \hat{\mu}_{Hst} + \hat{p}_{Hst} - \hat{p}_{Cst} + \hat{y}_s \right), \quad (66)$$

where κ_s are positive weights corresponding to consumption resources from each sector at baseline.

The expression analogous to equation (59) is now:

$$\hat{y}_s = \frac{\alpha_s}{1 - \alpha_s} \left(\hat{p}_{Hst} - \sum_{s'} \frac{\alpha_{s's}}{\alpha_s} \hat{p}_{Ms'st} - \hat{\mu}_{Hst} \right) \quad (67)$$

where $\sum_{s'} \alpha_{s's} = \alpha_s$. This follows a logic similar to that presented earlier, where markups of the sector are hindering production. However, note that in the multi-sector case inputs used are no longer priced the same as output.

I can also find a multi-sector version of equation (61):

$$\hat{c}_t = \sum_s \kappa_s \left(J_s \hat{\mu}_{Hst} + \hat{p}_{Hst} - \hat{p}_{Cst} + \frac{\alpha_s}{1 - \alpha_s} \left(\hat{p}_{Hst} - \sum_{s'} \frac{\alpha_{s's}}{\alpha_s} \hat{p}_{Ms'st} \right) \right) \quad (68)$$

with $J_s < 0$. This expression shows how variable markups enhance the gains from trade even in the multi-sector case, driven by the increase in efficiency. This, however, does not guarantee that all markups will contribute towards higher gains, as, for example, sectors receiving less expensive inputs will increase their markups and detract from welfare.

Combined, these examples provide the main intuition of the model: The competitive effects of imports increase the gains from trade. This intuition is true in the one-sector and the multi-sector models. However, the previous analysis required some simplifying assumptions, and is based on log-linear approximations. To confirm that my intuition holds in general equilibrium, I present my results in section 4.

4 Results

In this section, I present the results of computing the model presented in Section 2. Before any results, I discuss the calibration strategy adopted to set the parameters, using the year 2007 as baseline. With the parameters set, I conduct two sets of exercises. First, to present the behavior of the model, I will set an exogenous shock in foreign marginal costs, once for consumption and once for inputs. Second, I will use my baseline calibration to retrieve the marginal costs that matches data on domestic expenditure shares, and use them to compare the change in U.S. welfare attributable to the increase in imports between 1997 and 2007 under different pass-through assumptions.

4.1 Calibration

The multi-sector structure means that some parameters are unique, others are vectors where each value corresponds to a sector, and others are matrices where the elements correspond to sector-by-sector parameters. In this sense, the dimension of the calibration increases depending on the number of sectors S . As a starting point, I set the number of sectors $S = 2$. There are 14 groups of parameters to be calibrated and 6 exogenous groups of variables, comprising $5 + 11S + 4S^2$ parameters and variable values, as listed below.

4.1.1 External Calibration

The first 8 sets of parameters are calibrated externally following preceding literature, and most will remain fixed throughout. The external calibration is summarized in Table 1.

Table 1: External Calibration

Definition	Value	Source
Decreasing returns to consumption	$\rho = 1$	Log-utility
CES across sectors s Consumption	$\vartheta = 1$	Cobb-Douglas
CES across sectors s' Inputs	$\kappa_s = 1$	Cobb-Douglas
Kimball coefficient (levels)	$\sigma_s = 3$	Comin and Johnson [2022]
Kimball coefficient (super-elasticity)	$\epsilon_s = 2$	Comin and Johnson [2022]
Inverse of the Frisch elasticity	$\psi = 2$	Chetty et al. [2011]
CES across sources Exports	$\epsilon_X = 3$	$\mu_X = \frac{\epsilon_X}{\epsilon_X - 1} = 1.5$
CES across varieties Exports	$\eta_{Xs} = 3$	Feenstra et al. [2018]

I start by discussing the parameters σ_s and ϵ_s of the Kimball aggregators, which determine the dynamic of markups through two pairs of objects. As evidenced in equations (52 – 57), the pair (σ_s, ϵ_s) determines markups μ_{Hst} , the elasticity of demand ε_{Hst} , the super-elasticity of demand Γ_{Hst} , and price-cost pass-through Φ_{Hst} . In a symmetric steady state, that is when $\frac{P_{Hs0}}{P_{Cs0}} = 1$, the demand index simplifies to $D_{Cs0} = \frac{\sigma_s - 1}{\sigma_s}$. Then the symmetric elasticity of demand becomes fixed at $\epsilon_{Hs0}^C = \sigma_s$, the super-elasticity is $\Gamma_{Hs0}^C = \frac{\epsilon_s}{\sigma_s - 1}$, and the cost pass-through is $\Phi_{Hst}^C = \frac{\sigma_s - 1}{\sigma_s - 1 + \epsilon_s}$. To be clear, this symmetry only occurs when all domestic and all foreign firms charge the same prices.

For my baseline, I set the same values for the pair across sectors $\{\sigma = 3, \epsilon = 2\}$ following [Comin and Johnson \[2022\]](#). These values give markups of $\mu_{Hst}^C = 1.5$ and Home pass-through of $\Phi_{Hst} = 0.5$. I also try using $\{\sigma = 2, \epsilon = 1\}$ as in [Gopinath et al. \[2020\]](#), with similar results (not reported). Other values used in the literature include $\{\sigma = 5, \epsilon = 4\}$ in [Gopinath and Itskhoki \[2010\]](#), $\{\sigma = 5, \epsilon = 10\}$ in [Smets and Wouters \[2007\]](#), and $\{\sigma = 5, \epsilon = 33\}$ in [Eichenbaum and Fisher \[2007\]](#). A deeper discussion of the implications of these parameters can be found in [Klenow and Willis \[2016\]](#). While discussing these parameters, note that when $\epsilon \rightarrow 0$, the Kimball aggregator simplifies to $\Upsilon(x) = x^{\frac{\sigma-1}{\sigma}}$, making markups fixed in all cases. This special case of the aggregator is equivalent to a nested CES structure with the same coefficient of substitution for each level.

In this line, I set the CES substitution coefficients for exports to $\eta_{Xs} = \epsilon_{Xs} = 3$, making markups of exports $\mu_{Xs} = 1.5$, the same as the (symmetric) domestic markups. Equating the two is consistent with the discussion on elasticities in [Feenstra et al. \[2018\]](#). For labor, I set $\psi = 2$ to match a Frisch elasticity of $\frac{1}{\psi} = 0.5$, as discussed in [Chetty et al. \[2011\]](#). Furthermore, I set $\rho = 1$, producing log utility in consumption, which is also standard in quasi-static models, where ρ is the rate at which the marginal utility of consumption decreases. This context does not include intertemporal decisions or risk. Finally, I set substitution parameters across sectors ϑ and κ_s to 1, so that sector composition follows a Cobb-Douglas structure. I keep the externally calibrated coefficients symmetric across sectors, so, for example, $\sigma_s = \sigma \forall s$.

4.1.2 Internal Calibration

The second batch for calibration consists of 6 sets of parameters and 2 sets of values, which are determined by matching 6 sets of moments in the data at baseline, and normalizing 2 sets. The moments from data are the domestic expenditure shares for consumption Λ_{s0}^C and for inputs $\Lambda_{s's0}^M$, the weight of inputs in total costs MS_{s0} , the weight of individual inputs in total costs $MS_{s's0}$, the sector shares of consumption CS_{s0} , and nominal GDP_{s0} . I will also assume no trade deficit $T_0 = 0$. These data moments pin down the parameters and values $(\nu_s, \xi_{s's}, \zeta_s, \alpha_s, \alpha_{s's}, Z_{s0}, T_{s0})$. When computing equilibrium off baseline, these parameters are held fixed at the baseline numbers. All data moments come from the input-output construction described below. The last two parameters, ς and C_s^* are set by normalizing domestic prices $P_{Ht} = W_t = 1$.

Table 2: Internal Calibration

Definition	Value	Target
Home bias Consumption	$\nu_s = \begin{pmatrix} 0.66 \\ 0.99 \end{pmatrix}$	Λ_{s0}^C
Home bias Inputs	$\xi_{s's} = \begin{pmatrix} 0.78 & 0.79 \\ 0.83 & 0.98 \end{pmatrix}$	$\Lambda_{s's0}^M$
Home bias Consumption sectors	$\zeta_s = \begin{pmatrix} 0.16 \\ 0.84 \end{pmatrix}$	CS_0
Inputs in Production	$\alpha_s = \begin{pmatrix} 0.79 \\ 0.50 \end{pmatrix}$	MS_{s0}
Inputs in Production from sector s'	$\alpha_{s's} = \begin{pmatrix} 0.43 & 0.10 \\ 0.36 & 0.40 \end{pmatrix}$	$MS_{s's0}$
Domestic Productivity	$Z_{s0} = \begin{pmatrix} 3.45 \\ 4.00 \end{pmatrix}$	GDP_{s0}
Labor dis-utility scale parameter	$\varsigma = 0.01$	$W_0 = 1$
Foreign Consumption	$C_{s0}^* = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$	$P_{Hs0} = 1$

I set my baseline in 2007 instead of 1997 for practical reasons. In particular, the home bias parameters for 1997 ν_s and $\xi_{s's}$ were sometimes very close to one, showing how closed the non-manufacturing sector was in 1997. This strong Home bias diminished precision, affecting results. Also, I am fixing nominal foreign transfers T_0 to prevent real transfers of resources.

4.1.3 Data for calibration

The internally calibrated parameters are set by matching moments in the model with moments in U.S. data. To retrieve analogous moments in the data, I construct an adjusted input-output table for the U.S. working with the national accounts from the Bureau of Economic Activity (BEA). More specifically, I combine the summarized tables (73 sectors) from 1997 to 2016 on Make, Use, and Import Matrices after re-definitions, transforming the

matrices in two dimensions.

First, for simplicity, I collapse the tables from 73 sectors to 2 sectors: manufacturing and non-manufacturing. The manufacturing sector is defined as all NAICS2 sectors of the manufacturing family, with the rest set as non-manufacturing. This division responds to the characteristics of the 1997-2007 period, in particular the liberalization of trade with China, which characterizes the period. The China Shock had a clear differential effect on manufacturing, so I separated manufacturing from the rest of the economy.

Second, the adjustments respond to limitations in the BEA data. For one, imports are not separately taken into account in the Make and Use tables, making it difficult to track down which inputs are domestic and which are imported. This tracking is necessary to retrieve the domestic expenditure shares. Second, the Make and Use tables are not industry-by-industry tables, which complicates the analysis of upstream and downstream effects. I combine the three tables to make a unified input-output matrix, following the derivation procedure for the total requirement tables mixed with the definition of the import tables.

The end product is a matrix matching industries to industries that also separately tracks domestic and foreign production. This adjusted input-output tables from 1997 to 2016 now comprises many variables present in the model, from which I construct the data moments used in the internal calibration, in particular:

$$\overline{GDP}_{st} = \overline{P_{Hst}C_{Hst}} + \sum_{s'} \overline{P_{Hs't}M_{Hs'st}} + \overline{P_{Xs}X_s} - \overline{P_{Mst}M_{st}} \quad (69)$$

$$\overline{CS}_{st} = \frac{\overline{P_{Cst}C_{st}}}{\overline{P_{Ct}C_t}} \quad (70)$$

$$\overline{MS}_{st} = \frac{\overline{P_{Mst}M_{st}}}{\overline{P_{Mst}M_{st}} + \overline{W_tL_{st}}} \quad (71)$$

$$\overline{MS}_{s'st} = \frac{\overline{P_{Ms'st}M_{s'st}}}{\overline{P_{Mst}M_{st}} + \overline{W_tL_{st}}} \quad (72)$$

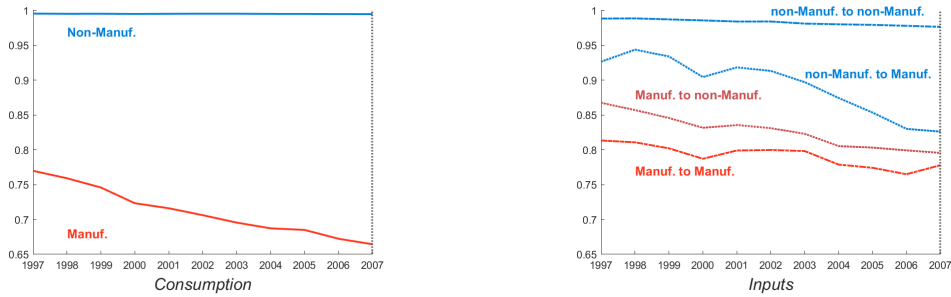
$$\overline{\Lambda}_{st}^C = \frac{\overline{P_{Hst}C_{Hst}}}{\overline{P_{Cst}C_{st}}} \quad (73)$$

$$\overline{\Lambda}_{s'st}^M = \frac{\overline{P_{MHs'st}M_{Hs'st}}}{\overline{P_{Ms'st}M_{s'st}}} \quad (74)$$

where $\overline{GDP}_{st}$ is the gross domestic product of sector s . $\overline{CS}_{st}$ is the share of consumption from sector s in aggregate consumption, $\overline{MS}_{st}$ is the share of all intermediate inputs in costs for sector s , while $\overline{MS}_{s'st}$ is the share of intermediate inputs from sector s' in costs for sector s . $\overline{\Lambda}_{st}^C$ is the domestic expenditure share of consumption in sector s , and $\overline{\Lambda}_{s'st}^M$ is the domestic expenditure share of inputs from sector s' bought by sector s . The line over the variables denotes data or transformations of data; note that the input-output tables are set in millions of dollars.

The domestic expenditure shares $\overline{\Lambda}_{st}^C$ and $\overline{\Lambda}_{s'st}^M$ are of particular importance in the literature, as [Arkolakis et al. \[2012\]](#) and [Arkolakis et al. \[2019\]](#) propose that they are a sufficient statistic for the increase in welfare for a large class of models. Therefore, I plot the domestic expenditure share of both consumption and inputs in Figure 1 below. It is clear from the figure that between 1997 and 2007, foreign manufacturing gained ground on domestic expenditure, in both consumption and inputs, as shown by the lines in red. Foreign exposure is a bit different for non-manufacturing sectors. The consumption of non-manufacturing goods remains almost entirely domestic, while non-manufacturing domestic input use by the own non-manufacturing sector drops only slightly. The remaining case does show a reduction in non-manufacturing inputs used to produce manufacturing goods.

Figure 1: Domestic Sourcing Shares



Input use is relevant to interpret this figure, as it is different across the two sector groups. These groups are evidenced in matrix A^{2007} below, the direct requirement matrix. Here, each element represents the weight of inputs from each row sector in total costs of the

column sector. For example, 0.43 is the weight of manufacturing inputs in manufacturing costs. The sum of each column is total input intensity, 0.79 for manufacturing and 0.50 for non-manufacturing, and thus the width of the cost channel. The off-diagonal elements suggest that the anti-competitive effects would be more potent from non-manufacturing to manufacturing $A_{(2,1)}^{2007} = 0.36$ than from manufacturing to non-manufacturing $A_{(1,2)}^{1997} = 0.10$.

$$A^{1997} = \begin{pmatrix} 0.43 & 0.10 \\ 0.36 & 0.40 \end{pmatrix} \quad (75)$$

4.2 Results - Exogenous Shocks

The first step is to compute the equilibrium for U.S. data in 2007 and use it as a benchmark. From this benchmark, and keeping the remaining parameters fixed, I introduce one of two reductions in foreign marginal costs. To avoid repetition as much as possible, I also sometimes refer to manufacturing as “sector 1” and non-manufacturing as “sector 2”.

First, to mimic import competition in manufacturing sales, I simulate a 10% reduction in the foreign marginal cost of manufacturing consumption goods, MC_{C1}^* . With a 50% pass-through, this translates to a 5% decrease in foreign prices. I shock foreign marginal costs instead of foreign prices because foreign prices are an equilibrium object, subject to their own variable markups. I report the corresponding results in the first columns of tables (3 – 5). Second, by inducing import competition on inputs, I lower marginal costs on the foreign manufacturing inputs used by the domestic non-manufacturing sector, MC_{M12}^* . The corresponding results are reported in the second columns of tables (3 – 5).

With these new foreign marginal costs, I recompute the equilibrium of the model and compare it to the benchmark. I present these results as percent changes from the benchmark. To summarize, I present only changes for markups (by market and sector), (sector) output and factors of production, and aggregate variables.

I will have a different for each sector, so in this exercise, I have two, one for manufacturing

and one for non-manufacturing goods. Each of these markups is, in turn, a weighted average of the markups charged in the markets served by each good. So, for example, the manufacturing markup is a combination of the markups charged for manufacturing goods sold for consumption, those sold as inputs for manufacturing production, and those sold as inputs for non-manufacturing production. In this sense, the first two rows of Table 3 represent the average markup of each industry, while the remaining six rows show the markup set in each market.

The first column of Table 3 shows how markups change as a result of the lower foreign marginal costs. The lower costs in the foreign economy are imperfectly passed through to import prices, which compete with domestic production. This results in a -2.4% drop in the markups of manufacturing consumption goods.

In addition, the lower manufacturing prices also lower domestic costs, which leads to an increase in domestic markups. This increase is particularly visible in markets that use manufacturing more intensively, which coincidentally is the manufacturing production market. This is the driver of the 0.7% increase in markups in μ_{H11}^M and μ_{H12}^M . This result also highlights how the pro-competitive and anti-competitive effects operate as a result of even a single shock.

Table 3: Results - Markups by Sector

		10 % drop	
		MC_{C1}^*	MC_{M12}^*
Markups M	μ_{H1}	-0.7	-0.3
Markups NM	μ_{H2}	0.0	0.0
Markups Consumption M	μ_{H1}^C	-2.4	0.4
Markups Consumption NM	μ_{H2}^C	0.0	0.0
Markups Inputs M to M	μ_{H11}^M	0.7	0.3
Markups Inputs M to NM	μ_{H12}^M	0.7	-1.9
Markups Inputs NM to M	μ_{H21}^M	0.1	0.0
Markups Inputs NM to NM	μ_{H21}^M	0.0	0.0
Simulated 10% drop in Foreign Marginal Costs			
Results in percent change from 1997 benchmark			

The second column of Table 3 shows the effect on markups of decreasing the foreign marginal costs of manufacturing goods sold to sector 2, non-manufacturing production. Similar to the previous case, the sharpest impact is on the domestic markups of competing input providers, “Inputs NM to M”, which decreases -1.9% . But here, there are also multiple markup-increasing effects. For example, markups for manufacturing consumption go up 0.4% .

Table 3 shows how different drops in foreign costs, which translate to lower foreign prices, impact domestic markups differently. Depending on how they relate to the exposed sector, markups will decrease when competing with imports or increase when the products are used as inputs.

Now, turning towards supply in each sector, Table 4 presents how production and factors

of production change in each sector. Starting with the first column, production in the exposed sector goes down -0.8% after the drop in foreign marginal costs. At the same time, production in non-manufacturing increases as a combination of the lower markups in manufacturing and the reallocation of labor.

Table 4: Results - Reallocation by Sector

		10 % drop	
		MC_{C1}^*	MC_{M12}^*
Labor M	L_1	-0.7	-0.3
Inputs M	M_1	-0.8	-0.1
Home Output M	Y_1	-0.8	-0.1
Labor NM	L_2	0.1	0.1
Inputs NM	M_2	0.1	0.8
Home Output NM	Y_2	0.1	0.4
Simulated 10% drop in Foreign Marginal Costs			
Results in percent change from 1997 benchmark			

The second column of Table 4 is also informative. Here, the price will drop for inputs bought by the non-manufacturing sector, creating both an increase in production due to better marginal costs and a reallocation of factors towards inputs. The net effect is a 0.4% increase in production.

Finally, Table 5 gives the aggregate results of each shock. Comparing results across the first line, the shock to consumption in sector 1 creates more welfare than the shock to manufacturing inputs used in sector 2. It also creates more jobs, and induces higher real wages. However, moving to the second half of the table, the shock on consumption seems

to improve the real wage bill by more, whereas the shock to inputs creates a larger gain in real profits. This is not a surprising result given that the shock in the first column lowers markups by more, whereas the shock in the second column has larger output.

Table 5: Results - Aggregate

		10 % drop	
		MC_{C1}^*	MC_{M12}^*
Consumption	C	0.31	0.26
Labor	L	0.07	0.03
Real Wages	$\frac{W}{P_C}$	0.45	0.31
Real GDP	$\frac{WL+\Pi}{P_C}$	0.31	0.26
Real Wage Bill	$\frac{WL}{P_C}$	0.52	0.34
Real Profits	$\frac{\Pi}{P_C}$	0.12	0.18
Simulated 10% drop in Foreign Marginal Costs			
Results in percent change from 1997 benchmark			

This exercise illustrates how different individual shocks are incorporated into the equilibrium, affecting markups, production, and welfare. However, these do not match any real-world change or moment in data. In the next exercise, I will do just that: I will retrieve the foreign marginal costs from the data and feed them through various specifications.

4.3 Results - Model Inversion

The objective of this exercise is to capture how changes in import competition affect welfare in my model. The first step is calibrating to the benchmark in year 2007. I will capture

import competition using data on the domestic expenditure shares in consumption and inputs for each sector Λ_{st}^C and $\Lambda_{s'st}^M$. Then I will compare results from this benchmark to other equilibria.

To be clear, I will take the ratio between my benchmark 2007 equilibrium in the denominator, and in the numerator, I will have 1997 equilibria under three different specifications. Comparing these ratios is appropriate across specifications because I assume a symmetric baseline calibration, meaning that the 2007 benchmark will have the same equilibrium under the three specifications.

I find the first off-benchmark equilibrium by internally calibrating marginal costs to 1997 data, taking the parameters calibrated to 2007 as given. I label this case “Base Pass-Through”, as it uses the same 50% pass-through as used in the benchmark calibration. In addition to computing the impact of import competition, this strategy also retrieves the $S + S^2$ foreign marginal costs that match the entry of foreign goods $\{MC_{CFst}, MC_{MFs'st}\}$. I call those retrieved foreign marginal costs the “inverted shocks”.

The second off-benchmark equilibrium consists of feeding those inverted shocks through the model, with one variation. The calibration for this exercise assumes most of the same parameters from the “Base Pass-Through” case, with the exception of the Kimball super-elasticity parameter which I now set to $\epsilon_s = \frac{1}{10}$. This alternate calibration keeps symmetric average markups the same in the benchmark but increases the cost pass-through to 95%. I label this the “High Pass-Through” case, and it delivers an effect characterized by the same marginal cost change with less responsive markups.

Finally, I compute a third off-benchmark equilibrium, now feeding the inverted shocks into an analogous model, replacing Kimball preferences and technology aggregation with the more common CES preferences and technology. As mentioned in the calibration section, this is a limiting case of the High Pass-Through case, where now $\epsilon \rightarrow 0$, so I label it the “Full Pass-Through” case.

The results are presented through a selection of variables in tables 5 – 8. These results

are shown as percent changes from 1997, analogous to the previous exercises. Results in the first column correspond to the Base Pass-Through equilibrium, where I compare equilibria in 1997 and 2007 with a 50% pass-through. In the second column I present results for the second off-baseline case, feeding the inverted shocks in the same model but now with a 95% pass-through. Finally, results in the third column present results using a 100% pass-through, the CES version of the model. Together, these exercises allow me to assess the role of variable markups, as ϵ_s controls how variable markups are. This super-elasticity of demand is also directly related to pass-through, as discussed before.

Table 6 below presents the markups in each exercise. Starting with the Base Pass-Through case, there is a generalized reduction in markups as the elasticity of demand changes in all markets. This is a net effect, as the countervailing forces displayed in the results for exogenous shocks are still operative. Markups in consumption manufacturing decrease the most, being -7.6% lower than 1997. In the High Pass-Through case, markups reasonably react by less, with the larger effect being a -0.5% in consumption manufacturing. Note that the Full Pass-Through case corresponds to fixed markups, so there is no change in markups with respect to 1997.

Table 6: Results - Markups by Sector

		Pass-Through		
		Base	High	Full
Markups M	μ_{H1}	-5.5	-0.4	—
Markups NM	μ_{H2}	-0.4	-0.0	—
Markups Consumption M	μ_{H1}^C	-7.6	-0.5	—
Markups Consumption NM	μ_{H2}^C	-0.0	-0.0	—
Markups Inputs M to M	μ_{H11}^M	-2.2	-0.2	—
Markups Inputs M to NM	μ_{H12}^M	-5.0	-0.3	—
Markups Inputs NM to M	μ_{H21}^M	-4.7	-0.3	—
Markups Inputs NM to NM	μ_{H21}^M	-0.5	-0.0	—
Matching 2007 change in import exposure				
Results in percent change from 1997 benchmark				

The productive reallocation from differential exposures to import competition is better appreciated in Table 7. I separate each sector as before and order them from factors of production L_j , M_j to output Y_j . In the Base Pass-Through case, we see both sectors grow with respect to 1997, with a reallocation of factors from labor towards intermediate inputs. This result is consistent with both the decrease in markups and the decrease in marginal costs of inputs stemming from import competition and gains through the cost channel.

Table 7: Results - Reallocation by Sector

		Pass-Through		
		Base	High	Full
Labor M	L_1	-2.8	-12.0	-12.6
Inputs M	M_1	11.6	-1.5	-2.3
Home Output M	Y_1	8.4	-3.8	-4.5
Labor NM	L_2	1.0	1.4	1.4
Inputs NM	M_2	9.1	7.4	7.3
Home Output NM	Y_2	5.0	4.4	4.3
Matching 2007 change in import exposure				
Results in percent change from 1997 benchmark				

The High and Full Pass-Through cases present a similar reallocation, with manufacturing output decreasing -3.8% and -4.5% respectively. The reduction is less marked on inputs, respectively -1.5% and -2.3% , and instead, the result is the sharp destruction of labor, -12.0% and -12.6% , respectively. The effects in non-manufacturing are a moderated version of the Base Pass-Through case, with slightly more growth in labor and slightly less growth in the use of intermediate inputs.

This result highlights the role of variable markups as a cushion for domestic production. The more variable markups are, the better domestic production fares. In the opposite direction, markups provide less of a cushion when the pass-through is higher.

Table 7 also shows a reduction in manufacturing labor across all specifications. This reduction is fueled by both changes in factor demand given the lower production and changes in favor of the more affordable input basket. This result is also consistent with the decline in manufacturing labor found in empirical literature of this period.

Before aggregating results, it is worth keeping track of sector sizes. Calibrating to the U.S. economy in 2007, manufacturing accounts for 19% of consumption and 14% of GDP. In that context, Table 8 shows that consumption grows 6.15% in the Base Pass-Through case, which I interpret as the gains from trade in this static model, which are net positive as expected. Labor grows 0.71% compared to 1997, and real wages increase 7.66%. In sum, real consumption, labor, and wages increase. Thinking on how consumption is financed, real GDP also increases by 6.15%, where by real GDP, I mean GDP over consumption prices. The wage bill grows by 8.43%, which is greater than the 4.13% increase in profits.

Table 8: Results - Aggregate

		Pass-Through		
		Base	High	Full
Consumption	C	6.15	5.17	5.14
Labor	L	0.71	0.27	0.24
Real Wages	$\frac{W}{P_C}$	7.66	5.75	5.65
Real GDP	$\frac{WL+\Pi}{P_C}$	6.15	5.17	5.14
Real Wage Bill	$\frac{WL}{P_C}$	8.43	6.04	5.91
Real Profits	$\frac{\Pi}{P_C}$	4.13	4.40	4.46
Matching 2007 change in import exposure				
Results in percent change from 1997 benchmark				

Here, the cases with higher cost pass-through have lower growth in real consumption of 5.17% and 5.14%, respectively. Also, more moderate changes occur in labor for each case, respectively 0.27% and 0.24%, and in real wages, 5.75% and 5.65%, resulting in a lower

growth for the wage bill of 6.04% and 5.91%. Somewhat surprisingly, real profits grow by more than in the Base Pass-Through case. This implies that both the growth and the size of non-manufacturing, combined with the relatively fixed markups, more than compensate for the decrease in manufacturing production.

All in all, the three comparisons demonstrate the role played by variable markups. The more variable the markups are, that is, the lower their pass-through, the higher the increase in real consumption. As for magnitudes, in my calibration, the base case increases consumption by 1% more than does the CES, which can be interpreted as reaping 20% more gains from trade.

The mechanism within the structure is also important. The pro-competitive and anti-competitive effects simultaneously cushion any shocks received by domestic production while helping transmission across sectors through cost. The input-output structure provides reallocation within and across sectors, taking into account cost channels of different widths in the structure of production. Finally, the general equilibrium framework allows for changes in the labor supply and wages as the economy faces more import competition.

5 Conclusion

In this paper, I study how including pro-competitive and anti-competitive effects can change the gains from trade in a multi-sector small open economy model with trade and an input-output structure. Computing this model to match U.S. data from 1997 and 2007, I find that the gains from trade increase by 20% when including variable markups. My computations also show how the internal reallocation of demand and variable demand elasticity work through the input-output structure, and how incomplete pass-through plays a consequential role.

This paper is a first step in incorporating richer competitive effects in trade models but suffers from some immediate shortcomings. First, for simplicity, the structure chosen is just sufficient to include domestic anti-competitive effects. However, there is room for improvement, both by expanding the number of sectors and by matching data on sector markups. In addition, increasing the number of sectors would have qualitative implications, as it increases the importance of imperfect cost pass-through and double marginalization.

A second immediate limitation is shown by the low labor destruction in manufacturing. There is consensus in the literature that the China Shock, an important flow of imports during this period, destroyed labor in manufacturing, but in my preferred specification, manufacturing labor decreases by less than -3% . This might be due to the parameters used in the calibration, in particular, fixed sector productivity, but also to the nature of the firm in this model. My model has neither an entry cost nor a fixed cost of operations, so a drop in profitability is just a drop in transfers to the owners of the firm, and the size of each sector remains the same. In a similar sense, the lack of investment means there is no resource reallocation from less profitable to more profitable firms. Enriching the supply side of the firms could remove this limitation. This second limitation might also speak to the nature of the decline in manufacturing labor linked to the number of firms and exit.

Taking into account firm heterogeneity within sectors would also be an informative future extension, as heterogeneity in competition across markets adds another dimension of

the competitive mechanism. Something similar to the analysis made by [Edmond et al. \[2018\]](#) would help me complement the understanding of how firm heterogeneity affects the gains from trade in a setting not too far removed from mine. If both the pro-competitive effect and the cost pass-through affect welfare, gains from trade will depend on what markets are liberalized, how competitive these markets are, and how the production network is organized. In general, foreign entry into markets that are more competitive and/or closer to the consumer should decrease the prices of final goods by more, while entry into less competitive markets and/or farther from the consumer will increase successive markups by more.

Finally, a more refined version of this model could help bridge the approaches taken by [Arkolakis et al. \[2019\]](#) and [Baqae and Farhi \[2023\]](#). Here, competitive effects are welfare-improving as in [Arkolakis et al. \[2019\]](#), but departing from fixed markups is not sufficient, as the vertical relation between sectors is instrumental to obtaining anti-competitive effects. The converse argument can be made of the welfare analysis in [Baqae and Farhi \[2023\]](#). If flexible markups compound the effects of trade liberalization along the input-output structure, the net welfare effects would differ from those using fixed wedges.

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Appendix - Equilibrium Summary

Given the values for $\{MC_{CFst}, MC_{MFs'st}, Z_{st}, P_{Cst}^*, C_{st}^*\}$ and parameters $\{\rho, \psi, \vartheta, \varsigma, \epsilon_s, \sigma_s, \eta_{Xs}, \epsilon_{Xs}, \zeta_s, \kappa_s, \nu_s, \alpha_s, \xi_{s's}, \alpha_{s's}\}$, the equilibrium conditions pin prices $\{P_{Hst}, MC_{st}, P_{Ct}, P_{Cst}, P_{Mst}, P_{Ms'st}, P_{Xst}, W_t, P_{CFst}, P_{MFs'st}\}$ and allocations $\{L_t, L_{st}, C_t, C_{st}, C_{Hst}, C_{Fst}, D_{Cst}, M_{st}, M_{s'st}, M_{Hs'st}, M_{Fs'st}, D_{Ms'st}, X_{st}, Y_{st}, \Pi_t\}$ as determined by the following system of equations

$$C_t^{-\rho} \frac{W_t}{P_{Ct}} = \varsigma L_t^\psi \quad (76)$$

$$C_{st} = \zeta_s \left(\frac{P_{Cst}}{P_{Ct}} \right)^{-\vartheta} C_t \quad (77)$$

$$C_{Hst} = \nu_s \Psi \left(D_{Cst} \frac{P_{Hst}}{P_{Cst}} \right) C_{st} \quad (78)$$

$$C_{Fst} = (1 - \nu_s) \Psi \left(D_{Ct} \frac{P_{CFst}}{P_{Cst}} \right) C_{st} \quad (79)$$

$$1 = \nu_s \Upsilon \left(\frac{C_{Hst}}{\nu_s C_{st}} \right) + (1 - \nu_s) \Upsilon \left(\frac{C_{Fst}}{(1 - \nu_s) C_{st}} \right) \quad (80)$$

$$P_{Cst} = P_{Hst} \frac{C_{Hst}}{C_{st}} + P_{CFst} \frac{C_{Fst}}{C_{st}} \quad (81)$$

$$P_{Ct} = \left(\sum_s \zeta_s P_{Cst}^{1-\vartheta} \right)^{\frac{1}{1-\vartheta}} \quad (82)$$

$$P_{Ct} C_t = W_t L_t + \Pi_t + T_t \quad (83)$$

$$W_t L_{st} = (1 - \alpha_s) Y_{st} MC_{st} \quad (84)$$

$$P_{Mst} M_{st} = \alpha_s Y_{st} MC_{st} \quad (85)$$

$$MC_{st} = Z_{st}^{-1} (1 - \alpha_s)^{-(1-\alpha_s)} \alpha_s^{-\alpha_s} W_t^{(1-\alpha_s)} P_{Mst}^{\alpha_s} \quad (86)$$

$$P_{Ht} = \frac{\epsilon_{Hst}}{\epsilon_{Hst} - 1} MC_t \quad (87)$$

$$M_{s'st} = M_{st} \left(\frac{\alpha_{s's}}{\alpha_s} \right) \left(\frac{P_{Ms'st}}{P_{Mst}} \right)^{-\kappa_s} \quad (88)$$

$$M_{Hs'st} = \xi_{s's} \Psi \left(D_{Ms'st} \frac{P_{Hs't}}{P_{Ms'st}} \right) M_{s'st} \quad (89)$$

$$M_{Fs'st} = (1 - \xi_{s's}) \Psi \left(D_{Ms'st} \frac{P_{MFs'st}}{P_{Ms'st}} \right) M_{s'st} \quad (90)$$

$$1 = \xi_{s's} \Upsilon \left(\frac{M_{Hs'st}}{\xi_{s's} M_{s'st}} \right) + (1 - \xi_{s's}) \Upsilon \left(\frac{M_{Fs'st}}{(1 - \xi_{s's}) M_{s'st}} \right) \quad (91)$$

$$P_{Ms'st} = P_{Hs't} \frac{M_{Hs'st}}{M_{s'st}} + P_{MFs'st} \frac{M_{Fs'st}}{M_{s'st}} \quad (92)$$

$$P_{Mst} = \left(\sum_{s'} \left(\frac{\alpha_{s's}}{\alpha_s} \right) P_{Ms'st}^{1-\kappa_s} \right)^{\frac{1}{1-\kappa_s}} \quad (93)$$

$$X_{st} = C_{st}^* \left(\frac{P_{Xst}}{P_{Cst}^*} \right)^{-\eta_{Xs}} \quad (94)$$

$$P_{Xst} = \frac{\epsilon_{Xs}}{\epsilon_{Xs} - 1} M C_{st} \quad (95)$$

$$P_{CFst} = \frac{\epsilon_{Fst}}{\epsilon_{Fst} - 1} M C_{Cst} \quad (96)$$

$$P_{MFs'st} = \frac{\epsilon_{Fst}}{\epsilon_{Fst} - 1} M C_{Ms'st} \quad (97)$$

$$Y_{st} = C_{Hst} + \sum_{s'} M_{Hss't} + X_{st} \quad (98)$$

$$L_t = \sum_s L_{st} \quad (99)$$

$$\Pi_t = \sum_s \left(\left(C_{Hst} + \sum_{s'} M_{Hss't} \right) P_{Hst} \frac{1}{\epsilon_{Hst}} + X_{st} P_{Xst} \frac{1}{\epsilon_{Xs}} \right) \quad (100)$$